

Horizontal Transmission of *Beauveria bassiana* in *Anastrepha ludens* (Diptera: Tephritidae) Under Laboratory and Field Cage Conditions

JORGE TOLEDO,¹ SERGIO E. CAMPOS,² SALVADOR FLORES,² PABLO LIEDO,¹
JUAN F. BARRERA,¹ ANTONIO VILLASEÑOR,² AND PABLO MONTOYA^{2,3}

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ABSTRACT The virulence of two products of the fungus *Beauveria bassiana* (LCPP and Bassianil) on adult *Anastrepha ludens* (Loew) (Diptera: Tephritidae) and their effect on the mating performance of infected males was evaluated in laboratory and field cage tests. The horizontal transmission capacity of the fungus during copulation or attempted copulation also was quantified using inoculated males as well as the impact of infection on female fecundity and longevity. Both fungal products were found to be highly virulent (LCPP, mortality = 98.7% at 1×10^8 conidia per ml, $LT_{50} = 4.20$ d, $LC_{50} = 9.35 \times 10^5$ conidia per ml; Bassianil, mortality = 99.3% at 1×10^8 conidia per ml, $LT_{50} = 4.04$ d, $LC_{50} = 2.69 \times 10^7$ conidia per ml). Mating success of inoculated males was not affected compared with the control group during the 3 d postinoculation. Horizontal transmission to females during the first day was 80.6 and 84.3% through mating and 15.4 and 21.6% through attempts to mate and contact during courtship for the LCPP and Bassianil products, respectively. The fertility of infected females was notably reduced, and longevity did not extend beyond 15 d. Our results suggest the possibility of using sterile flies as fungus vectors in sterile insect technique programs, but the potential benefits and shortcomings of this approach require further investigation.

KEY WORDS *Anastrepha ludens*, *Beauveria bassiana*, horizontal transmission, virulence, pathogenicity

The Mexican fruit fly, *Anastrepha ludens* (Loew) (Diptera: Tephritidae), is an important pest that limits fruit exports to countries that are free of this pest (Aluja 1994, APHIS 1994, NOM 1998). Integrated pest management (IPM) programs have been established at national and regional levels to establish pest-free areas (Reyes et al. 2000, Enkerlin 2005). The most common control methods used in these IPM programs have been the application of toxic bait sprays (e.g., malathion + hydrolyzed protein) and the sterile insect technique (SIT). Insecticidal bait sprays have been effective (Dowell et al. 2000), but they can generate environmental and social concern given their broad spectrum of activity and the large areas over which they are applied (Troetschler 1983, Michaud 2003).

Biological control by using entomopathogens such as the fungus *Beauveria bassiana* (Bals.) Vuill., could represent a suitable alternative (De la Rosa et al. 2002). The effectiveness of *B. bassiana* has been demonstrated in several species of Diptera, including the house fly, *Musca domestica* L. (Barson et al. 1994; Watson et al. 1995, 1996); tsetse flies (*Glossina* spp.), (Kaaya and Munyinyi 1995, Maniania and Odulaja

1998); and some fruit flies, such as *A. ludens*; the Mediterranean fruit fly, *Ceratitis capitata* (Wiedemann), and other species of the genus *Ceratitis* (Castillo et al. 2000, De la Rosa et al. 2002, Ekesi et al. 2002).

However, because *B. bassiana* is a natural enemy of a wide range of insect species (Alves and Lecuona 1996), cover sprays of this fungus could affect nontarget organisms. To minimize these potential adverse effects, we explore here the feasibility of using sterile males as vectors with the aim of reducing the amount applied and area treated with the fungus, and, possibly, increasing the effectiveness of sterile insects in IPM programs.

The potential for horizontal transmission of this fungus has been demonstrated by Kreutz et al. (2004) and Meadow et al. (2000) who showed that infected individuals could serve as efficient vectors to transmit the infection to untreated individuals in *Ips typographus* (L.) (Coleoptera: Scolitidae) and *Delia radicum* (L.) (Diptera: Anthomyiidae), respectively.

Our goals in this study were to 1) evaluate the virulence of two products formulated with conidia of *B. bassiana* on adult *A. ludens*; 2) determine the effect of these products on the mating performance of infected sterile males; and 3) examine the efficiency of horizontal transmission from infected males to healthy females as well as the impact of the infection on female fecundity, fertility, and longevity.

¹ Departamento de Entomología Tropical, El Colegio de la Frontera Sur, Apartado Postal 36, Tapachula, Chiapas, 30700 Mexico.

² Subdirección de Desarrollo de Métodos, Programa Moscamed, SAGARPA, Central Poniente 14, Tapachula, Chiapas, 30700 Mexico.

³ Corresponding author, e-mail: pmontoya@prodigy.net.mx.

Materials and Methods

Laboratory and field cage experiments were carried out at ECOSUR facilities in Tapachula and Moscamed facilities (DGSV-SAGARPA) at Metapa de Domínguez, Chiapas, Mexico.

Biological Material. *A. ludens* adults were obtained from the Moscafrut facility, where they were mass reared as described by Domínguez et al. (2000).

One of the *B. bassiana* products selected for this study was obtained from the Entomopathogens Production Laboratory of the Local Committee on Plant Protection (LCPP, 3.5×10^9 conidia per g), located in Tapachula, Mexico, whereas the other product (Bassianil, 2×10^9 conidia per g) was provided by Biotropic Inc. (Guadalajara, Mexico). Both products were stored at 4°C for 1 mo before used. The viability of the conidia of both products was assessed before the experiments by using the microculture method described by Jiménez (1992). Conidia viability was >95% for both products. Laboratory environmental conditions were $27 \pm 2^\circ\text{C}$, $80 \pm 5\%$ RH, and a photoperiod of 12:12 (L:D) h.

Virulence Test. A stock suspension of *B. bassiana* was prepared by suspending 1 g of conidia in 100 ml of sterile distilled water mixed with 0.7% of glycerin as dispersant in a glass flask, and homogenized with a magnetic agitator for 30 min. The median lethal time (LT_{50}) and median lethal concentration (LC_{50}) were calculated by exposing adult *A. ludens* aged 4–6 d to different suspensions of *B. bassiana*: 1×10^5 , 5×10^5 , 1×10^6 , 5×10^6 , 1×10^7 , 5×10^7 , 1×10^8 conidia per ml, and a control of sterile distilled water. Each group of flies was treated by immersion in 1.5 ml of conidia suspension for 30 s, with gentle agitation. The treated flies were placed in petri dishes with filter paper to remove any excess moisture, and they were then transferred to clear 8-cm-diameter by 12.5-cm-height clear plastic containers. The flies were fed with a mixture of hydrolyzed yeast (MP Biomedicals, Irvine, CA) and sucrose in a 1:3 ratio. Water was supplied in vials with cotton wicks. Mortality was recorded daily, over a period of 10 d or until the last fly died. To determine whether the insects died from fungal infection, dead flies were placed in moist chambers to stimulate mycelia growth and sporulation. Each bioassay was repeated five times by using five *A. ludens* males and five females per concentration.

Horizontal Transmission of Conidia in Field Cages. Two thousand *A. ludens* pupae were allowed to emerge in 27-dm³ glass cages (30 by 30 by 30 cm), one side of which was covered with nylon gauze. Once the adults had emerged, they were separated by sex and placed in separate cages, where they were provided with food (hydrolyzed yeast and sucrose in a 1:3 ratio) and water until they reached sexual maturity (9 d after adult emergence).

One hundred sexually mature virgin males were treated with conidia from each bioinsecticide product. To facilitate their management during the inoculation process, insects were placed in test tubes and they were placed in a refrigerator at 3°C for 5 min to render

them lethargic. Cooled flies were then placed in groups of 50 individuals in petri dishes containing 0.5 g of conidia, and they were gently shaken for 1 min. One of the following treatments was applied to males: 1) males inoculated with Bassianil, 2) males inoculated with the LCPP product, and 3) untreated males (control group). Treated insects were placed into a 70- by 70- by 70-cm wire cage, lined with nylon netting with access to food and water until required for a field cage experiment.

Nine days later, the field cage experiment was carried out. Four field cages (2.20 m in height by 3.0 m in diameter) were used over a period of four consecutive days. A small mango, *Mangifera indica* L., tree (120 cm in height) was placed in the center of each field cage. The cages remained outdoors under the shade of *Inga* sp. and tropical almond, *Terminalia catappa* L., trees. Average temperature was $25.9 \pm 1.1^\circ\text{C}$, and average relative humidity ranged from 43 to 97% during the experimental period. Treatments were randomly assigned to field cages.

Given that *A. ludens* courtship and mating behavior occurs at dusk (Aluja 1994), observations were performed between 1600 and 2000 hours. In each cage, 100 treated males and 100 untreated females were released. Flies were continuously observed and when mating occurred each couple was captured in a 15- by 45-mm vial. The number of matings as well as the duration of each copula was recorded. After mating, the flies were placed individually into 8-cm-diameter by 12.5-cm-height clear plastic containers with food and water, and daily mortality was recorded over a period of 20 consecutive days.

To determine the prevalence of transmission to flies that had not mated (and were not recovered from the field cages), the remaining insects were captured the next day and taken to the laboratory to quantify mortality due to fungal infection. These flies were placed in 27-dm³ glass cages with food and water, and daily mortality was recorded over a 20-d period. Dead flies were placed in a moist chamber, and the number that developed signs of sporulation was recorded. Four replicates of this experiment were done.

Survival, Fecundity, and Fertility of Infected Females. Under laboratory conditions, 9 d after adult emergence, pairs of virgin *A. ludens* were placed in 1-dm³ (10- by 10- by 10-cm) cages with food (protein/sugar, 1:3) and water. The following treatments were performed using the LCPP product: 1) noninfected female + infected male and 2) noninfected female + noninfected male (control). An oviposition device consisting of a green agar sphere wrapped in Parafilm and measuring 2 cm in diameter (Boller 1968, Freeman and Carey 1990) was placed in each cage and changed daily to record the number of eggs laid per female during their lifetime. To determine egg hatch, eggs extracted from the agar devices were lined up on a piece of moist black fabric placed on a water saturated sponge following the method described in the SIT quality control manual for tephritid flies (FAO/IAEA/USDA 2003). Egg hatch was recorded after 5 d of incubation. Fifteen replicates were done.

Parameter	Transmission to females having mated			<i>P</i>	Transmission to females without mating			<i>P</i>
	LCPP	Bassianil	Control		LCPP	Bassianil	Control	
No. of matings	23.9 ± 5.1	17.4 ± 4.8	26.6 ± 5.1	$F_{2, 35}; P > 0.05$				
Copula duration (min)	79.2 ± 2.9	73.6 ± 2.3	75.7 ± 2.2	$F_{2, 1084}; P > 0.05$				
Sporulation ♂ transmitter (%)	100.0	97.7		$\chi^2_{25}; P > 0.05$	91.4	92.0		$\chi^2_{25}; P > 0.05$
Sporulation ♀ receiver (%)	80.6	84.3		$\chi^2_{25}; P > 0.05$	15.4	21.6		$\chi^2_{25}; P > 0.05$
Longevity ♂ transmitter (d)	3.8 ± 0.3	4.2 ± 0.3	18.0 ± 0.3	$F_{2, 246}; P < 0.01$	3.4 ± 0.1	2.9 ± 0.1	17.1 ± 0.4	$F_{2, 297}; P < 0.01$
Longevity ♀ receiver (d)	7.2 ± 0.6	6.3 ± 0.5	14.7 ± 0.3	$F_{2, 224}; P < 0.01$	6.1 ± 0.6	5.3 ± 0.5	17.9 ± 0.4	$F_{2, 460}; P < 0.01$

Statistical Analysis. The mortality results from the laboratory bioassays were subjected to Probit analysis (SAS Institute 2002). Lethal time (LT) and lethal concentration (LC) values were compared using their respective 95% confidence limits (CLs) for each fungal product. Mating periods were compared by log-rank analysis. The number of matings, mortality, fecundity and fertility were subjected to analysis of variance (ANOVA) followed by means separation using the Tukey test ($P \leq 0.05$) (SAS Institute 2002). Percentage of fertility values were arcsine transformed, whereas other parameters were $\log(x + 1)$ transformed to comply with ANOVA assumptions. The prevalence of transmission of the fungus via mating and attempted mating was compared using a contingency table (SAS Institute 2002). Survival and fecundity graphs were constructed following Carey (1993).

Results

Virulence Test. There was no mortality in control insects during the bioassay period. Both *B. bassiana* products were highly virulent to adult *A. ludens*. At a concentration of 1×10^8 conidia per ml for the LCPP product, a mortality of 98.7% was observed along with a LT_{50} (95% confidence interval [CI]) of 4.20 (3.99–4.97) days. The same concentration of Bassianil re-

sulted in 99.3% mortality and an LT_{50} (95% CI) of 4.04 (3.73–4.37) days. The LC_{50} (95% CI) values were estimated to be 9.35×10^5 (6.66×10^5 – 1.32×10^6) and 2.69×10^7 (1.76×10^7 – 4.15×10^7) conidia per ml for the LCPP and Bassianil products, respectively, indicating that the LCPP product was significantly more virulent than Bassianil. The concentration–mortality regression for Bassianil showed a slope of 1.23 ± 0.01 (goodness-of-fit test: $\chi^2 = 1.64$, $P = 0.896$), whereas the regression slope of the LCPP product was 1.10 ± 0.01 (goodness-of-fit test: $\chi^2 = 5.85$, $P = 0.036$).

Horizontal Transmission of Conidia in Field Cages. The mean number of matings per infected male and non infected male was not significantly different ($F = 1.06$, $df = 2, 35$; $P = 0.359$). There was no significant difference in copula duration ($F = 1.12$; $df = 2, 1,084$; $P = 0.325$) (Table 1).

Mortality caused by *B. bassiana* in females that mated with infected males was confirmed by the prevalence of sporulation (Fig. 1). On the first day, fungal transmission by inoculated males was between 80.6 and 84.3% in females that had mated with LCPP- and Bassianil-treated males, respectively. For those females that did not mate but came into contact with males through lek courtship and mating attempts, fungal transmission was 15.4 and 21.6% for each product, respectively.

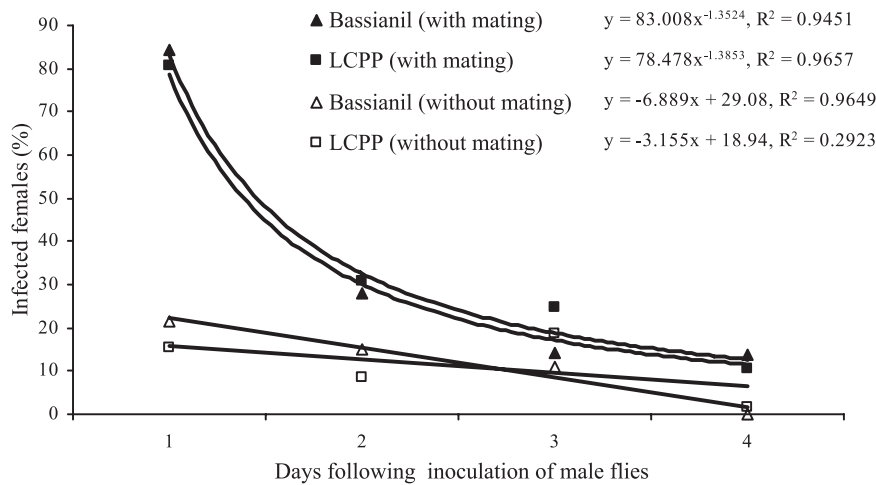


Fig. 1. Percentage of female *A. ludens* infected in field cages by contact (with and without mating) with males inoculated with two products of *B. bassiana*.

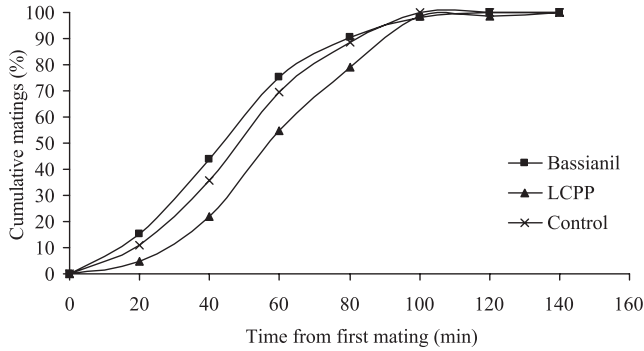


Fig. 2. Comparison of accumulated percentage of successful matings by groups of male *A. ludens* inoculated with two *B. bassiana* products and an untreated control group, over a period of 3 h.

The accumulated mating percentage by inoculated males was not significantly different from that of the control group (Fig. 2). This finding suggested that males suffered no adverse effects in their ability to mate during the 3 d after inoculation.

Survival, Fecundity, and Fertility of Infected Females. *A. ludens* females infected with *B. bassiana* through mating with infected males suffered a significant reduction in their mean fecundity (mx) in comparison with the control group (Fig. 3). Net fecundity was 85.07 ± 19.73 and $2,402 \pm 255.03$ eggs per female

for treated and control females, respectively ($F = 58.95$; $df = 1, 28$; $P < 0.001$). The daily fecundity was 36.5 ± 3.9 eggs per female per d for treated females and 52.6 ± 5.6 eggs per female per d for control females ($F = 5.91$; $df = 1, 33$; $P = 0.020$). Egg production lasted up to 80 d in the control group, whereas in infected females, egg production did not go beyond 15 d (Fig. 3). Fertility, expressed as the percentage of hatched eggs, was not affected by fungal infection. Mean egg hatch was $62.21 \pm 6.21\%$ for treated and $56.67 \pm 4.95\%$ for control females ($F = 0.68$; $df = 1, 25$; $P = 0.417$).

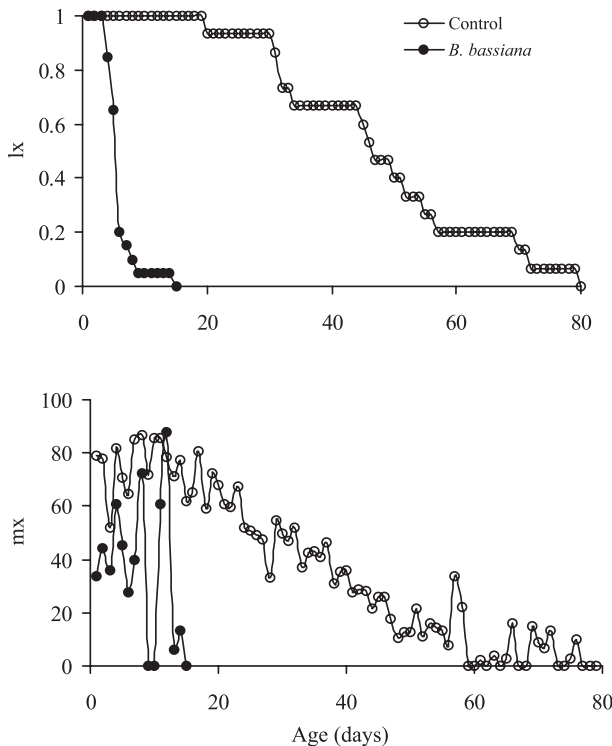


Fig. 3. Survival (lx) and fecundity (mx) of female *A. ludens* infected with *B. bassiana* (LCP product), through mating with experimentally inoculated males.

Discussion

Our results indicate that both the *B. bassiana* that we tested were pathogenic to *A. ludens* adults. This is in agreement with De la Rosa et al. (2002) and with studies on other fruit fly species. Dimbi et al. (2003) reported up to 100% mortality with *B. bassiana* in adult *C. capitata*, and in *C. rosa* variety *fasciventris* Muñoz (2000) tested 16 products of *B. bassiana* against *C. capitata* and observed between 20 and 98.7% mortality. The high virulence of these two products was confirmed by the LC₅₀ values calculated after laboratory bioassays.

Infected males were capable of transmitting the pathogen to uninfected females through copulation, that lasted for >1 h, and through mating attempts, that lasted only a few seconds. Studies with other species of Diptera report similar findings (Kaaya 1989). For example, horizontal transmission in species of Glossinidae infected with *B. bassiana* and *Metarhizium anisopliae* (Metsch.) Sorokin was observed under laboratory conditions (Kaaya and Okech 1990). Infected males were able to contaminate uninfected females, resulting in 65% mortality with *B. bassiana* and 55% with *M. anisopliae* after 32 d. In our study, the longevity of both the conidia donor males and the conidia receptor females was significantly reduced compared with that of control flies.

The reproductive capacity of males (e.g., number of matings and duration of copula) was not significantly affected by infection status during the first 3 d post-inoculation, when the highest probability of transmission to females occurred. This is the time when the infection process is taking place in male vectors (i.e., germination of conidia and penetration through the insect's cuticle). The infected insect dies just before the fungal invasion of the hemocoel is complete, but the time needed to die depends on the pathogen strain, insect host and environmental conditions (Lecuona et al. 1996). In our study, the *B. bassiana* products used provided the males with a temporal opportunity for courtship and mating.

The feasibility of horizontal transmission represents a new opportunity for IPM programs, by integrating biological control with the SIT. The concept of auto-dissemination, which uses contaminated insects to disperse a pathogen into a larger insect population, has been tested for a variety of pests insects, including Diptera. For example, tsetse flies were attracted to traps where they were infected with conidia of *M. anisopliae* by using an autoinoculative device (Maniania 1998).

Our findings here could be of particular relevance to SIT fruit fly control programs. Evidence from release-recapture experiments suggest that sterile flies live only a few days after they are released (Baker et al. 1986, Baker and Van der Valk 1992, Hendrichs et al. 1993, Thomas and Loera-Gallardo 1998). Because of selection under mass rearing conditions, they might be highly susceptible to predation (Hendrichs et al. 1993, 1994; Hendrichs and Hendrichs 1998). This trait will not be observed in longevity tests under laboratory

conditions. Assuming that sterile flies live very shortly under natural field conditions, the effectiveness of these short living sterile flies can be improved if they could be used as vectors or conidia transmitters. These sterile flies could have two possible effects on the wild population: the induction of sterility and increased mortality due to fungal dissemination. Fertility and longevity of females that mated with infected males was drastically reduced in comparison with control females. This suggests that the potential of infected females to infest fruit also could be reduced and could represent an additional advantage of this control strategy.

However, the effect of *B. bassiana* on the performance of the sterile insects needs to be investigated further. It will be necessary to determine whether fungal inoculation of sterile insects affects their ability to fly, to avoid predators, to find leks, to court, and to compete with wild males for matings. This will involve comparing the overall effectiveness of the conventional SIT with that of the SIT-pathogen combination. If sterile flies are living longer under current SIT programs conditions, and they are inducing sterility into the wild population later in their life, the SIT-pathogen approach could be detrimental.

We suggest that there are several potential advantages of this approach: 1) its specificity to the target pest; 2) the ability to place the pathogen conidia on the target pest individuals will likely reduce effects on nontarget organisms (Ekesi et al. 2005); and 3) the potential reduction of the proportion of wild females that remate (Orozco et al. 2001, Rull et al. 2005), thereby increasing SIT effectiveness. The possibility of male-male transmission through intrasexual competition, and the possibility of finding new fungal products with longer lethal times, might represent further improvements. We think that this novel SIT-pathogen approach deserves further investigation.

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